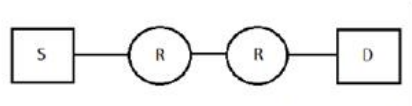


JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
COMPUTER NETWORKS & INTERNET of THINGS (18B11CS311)
BTECH 6TH SEM Even_2021
TUTORIAL-1 & 2 (Week 1-2) (18 January -30 January 2021)

1. Briefly describe how the Internet's connection-oriented service provides reliable transport.
2. What advantage does a circuit-switched network have over a packet-switched network?
3. A designer wants to design a point to point sub network with 10 routers of full duplex line. Then the total no of lines among them would be
 - A.10
 - B.20
 - C.45
 - D.90
4. A network designer wants to connect 5 routers as point-to-point simplex line. Then the total no. of lines required would be
 - A.5
 - B.10
 - C.20
 - D.32

5.

Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



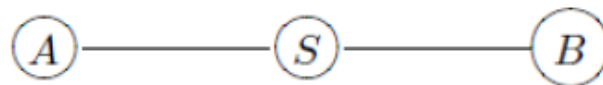
- (A) Network layer - 4 times and Data link layer - 4 times
 - (B) Network layer - 4 times and Data link layer - 3 times
 - (C) Network layer - 4 times and Data link layer - 6 times
 - (D) Network layer - 2 times and Data link layer - 6 times
6. Suppose that between a sending host and a receiving host there is exactly one packet switch. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. Assuming that the router uses store-and-forward packet switching, what is the total end-to-end delay to send a packet of length L . (Ignore queuing and propagation delay).
7. The number of point to point links required in a fully connected network for 50 entities is
 - A.1250
 - B.1225
 - C.2500
 - D. 50

8. When a host on network A sends a message to a host on network B, which address does the router look at?

- A) port
- B) logical
- C) physical
- D) none of the above

9.

Hosts A and B are each connected to a switch S via 10-Mbps links as shown above. The propagation delay on each link is $20\mu\text{s}$. S is a store-and-forward device that can send and receive bits simultaneously; it begins retransmitting a received packet $35\mu\text{s}$ after it has finished receiving it (if it can). Calculate the total time in milliseconds required to transmit 12,000 bits from host A to host B



10.

Consider an application that transmits data at a steady rate (for example, the sender generates an N-bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:

- a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?
- b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

11.

In this problem, we consider sending real-time voice from Host A to Host B over a packet switched network (VoIP). Host A converts analogue voice to a digital 64 kbps bit stream on the fly. Host A then groups the bits into 56-byte packets. There is one link between Hosts A and B; its transmission rate is 2 Mbps and its propagation delay is 10 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analogue signal. How much time elapses from the time a bit is created (from the original analogue signal at Host A) until the bit is decoded (as part of the analogue signal at Host B)?

12.

In modern packet-switched networks, including the Internet, the source host segments long, application-layer messages (for example, an image or a music file) into smaller packets and sends the packets into the network. The receiver then reassembles the packets back into the original message. We refer to this process as *message segmentation*. Figure 1.27 illustrates the end-to-end transport of a message with and without message segmentation. Consider a message that is $8 \cdot 10^6$ bits long that is to be sent from source to destination in Figure 1.27. Suppose each link in the figure is 2 Mbps. Ignore propagation, queuing, and processing delays.

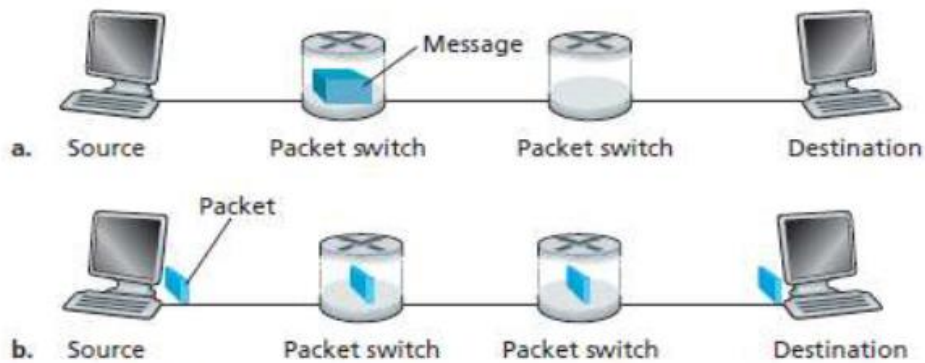


Figure 1.27 ♦ End-to-end message transport: (a) without message segmentation; (b) with message segmentation

- Consider sending the message from source to destination *without* message segmentation. How long does it take to move the message from the source host to the first packet switch? Keeping in mind that each switch uses store-and-forward packet switching, what is the total time to move the message from source host to destination host?
- Now suppose that the message is segmented into 800 packets, with each packet being 10,000 bits long. How long does it take to move the first packet from source host to the first switch? When the first packet is being sent from the first switch to the second switch, the second packet is being sent from the source host to the first switch. At what time will the second packet be fully received at the first switch?
- How long does it take to move the file from source host to destination host when message segmentation is used? Compare this result with your answer in part (a) and comment.
- In addition to reducing delay, what are reasons to use message segmentation?

13.

For the following, assume that no data compression is done, although in practice this would almost never be the case. For (a) to (c), calculate the bandwidth necessary for transmitting in real time:

- Video at a resolution of 640×480 , 3 bytes/pixel, 30 frames/second.
- Video at a resolution of 160×120 , 1 byte/pixel, 5 frames/second.
- CD-ROM music, assuming one CD holds 75 minutes' worth and takes 650 MB.
- Assume a fax transmits an 8×10 -inch black-and-white image at a resolution of 72 pixels per inch. How long would this take over a 14.4-kbps modem?

14. Consider a simple network as follows: $12P$ bits of data need to be sent from node A, through node B, to node C. Link A-b and link B-C are identical. The link bandwidth is R bps. The link speed is s mps. Each link is of length D meters. Suppose the packet header containing necessary information to deliver any amount of data from A to C is P bits. Consider queuing and processing delays are negligible.

- If we send all the $12P$ bits data in one packet, calculate the delay for all the data to arrive at C.
- Calculate the total delay for the data to arrive at C if the user data is sent in two equal sized packets.